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ZOO506-Applied-Entomology-And-Pest-Management PAST PAPERS

1. Order coeloptera with example. 10

Coleoptera

- **Includes: Beetles and Weevils**

1. Blister beetle
2. Boll weevil
3. Collops beetle
4. Darkling beetle
5. Dermestid beetle
6. Dung beetle
7. Fig beetle,
8. Flea beetle
9. Lady beetle
10. Long-horned beetle
11. Palo Verde root borer
12. Metallic wood borer

- **Mouthparts:** Chewing, Weevil- Piercing Sucking
- **Metamorphosis:** Complete Metamophosis
- **Beetles** are a group of insects that form the order **Coleoptera**, in the superorder Endopterygota. Their front pair of wings are hardened into wing-cases, elytra, distinguishing them from most other insects. The Coleoptera, with about 400,000 species, is the largest of all orders, constituting almost 40% of described insects and 25% of all known animal life-forms; new species are discovered frequently.

2. What things keep in mind during field scouting.5 marks

3. Realistically, a field scouting must know:
4. How to properly identify pests and their damage
5. How to get an accurate assessment of pest populations
6. A field scouts objective is to provide an accurate and unbiased objective of pest populations
7. ▶ What tools and procedures do you use for alfalfa weevil for potato leafhopper, etc.

8. Both of these examples are pests on alfalfa, however, the sampling procedure is different.

3. Order, family and symptoms of tobacco cutworm.5

Armyworm /Tobacco Cutworm: *Spodoptera litura*

- **Family:** Noctuidae ▶ **Order:** Lepidoptera.

- It is a major and important chewing insect pest of cotton crop.

Symptoms of damage

- Scrapping the epidermal layer, leaving the skeleton of veins of leaf
- During severe attack, only the stem and side shoots will be standing in the field without any leaf or bolls
- Larvae feed the leaves by making small holes.

4. Why foreign insects brought in the country.3

1. Various laws and regulations both Federal and state have been enacted to prevent the introduction of foreign pests:
    a. **To prevent their spread within the country**
    b. **To enforce control and extermination**
    c. **To ensure that chemicals used for controlling insects are neither adulterated nor misbranded**
2. Which in many instances have proved valuable in reducing the numbers of foreign pests
3. Probably the first introduction attempted from a foreign country was the now classic case of a **ladybeetle brought from Australia** to attack the **cottony cushion scale**
4. From abroad **to combat those foreign insect pests** brought to this country accidentally but without their natural enemies

5 Demage control by termies.3

1. The use of partially decomposed manure should be avoided ▶

Locate and destroy the termite colony near by field

- Irrigation water with crude oil emulsion.
- The removal of decaying organic matters cow dung, wood or dry stubbles from the field
- Setts treatment with Imidacloprid (0.1%) or Chlorpyriphos 20 EC 0.04 % for 5 min.
- Application of well rotten Neem cake manure @ 60 Cartloads/ ha.
- Spray Chlorpyriphos 20%EC 750 ml/ha

6 What is biological control 2 marks

Biological control or biocontrol is a method of controlling pests such as insects, mites, weeds and plant diseases using other organisms. It relies on predation, parasitism, herbivory, or other natural mechanisms, but typically also involves an active human management role.

7 Method of biological control.5

Biological control methods

Biological control involves the use of insects or pathogens (diseases) that affect the health of weeds (called invasive plants in the Biosecurity Act 2014). Usually, these biocontrol agents are from the same country of origin as the weed species.

Biosecurity Queensland undertakes biological control research in Queensland. Strict measures are in place to ensure that these agents do not negatively affect native plants and animals or horticultural and agricultural crops.

Biosecurity Queensland's biological control group is presently working on 10 weeds of concern to Queensland:

- bellyache bush
- cat's claw creeper
- lantana
- madeira vine
- mother-of-millions
- parthenium weed
- prickly acacia
- siam weed
- cabomba
- opuntioid cacti.

A number of biological controls have been released at strategic infestation sites, and are spreading to other infestations of their host plant.

Biological control agents can reduce the vigour, size and competitiveness of weed infestations; however, they rarely get rid of them altogether. Biological control works best in conjunction with other control methods

8 Appearance of adult rice-ear.2

Rice ear- head bug

▶ The adult of *Leptocorisa oratorius is green, light brown or* mixed yellow in color ▶

The adult are slender and about 20mm long.

9 Define arthropods? 2

An “arthropod” is an invertebrate animal that has an exoskeleton, a segmented body, and jointed appendages

any of a phylum (Arthropoda) of invertebrate animals (such as insects, arachnids, and crustaceans) that have a segmented body and jointed appendages, a usually chitinous exoskeleton molted at intervals, and a dorsal anterior brain connected to a ventral chain of ganglia

10 Damage by termites?3

Termite

- *SYMPTOMS OF DAMAGE*
- They enter through cut ends of setts & feed on the soft tissue.
- The tunnel is filled with the soil.
- The termites attack setts, shoots, canes and also stubbles ▶ Entire shoot dries up and can be pulled out easily

11 What are attractant? 3

Attractants

1. Sometimes called **attrahents** are materials used to **lure insects** to traps and poisoned baits
2. The use of geraniol in **Japanese beetle traps**, **sugar sirups** for **ant baits**, and **fermenting sirup solutions** for trapping various moths are good examples.

12 Advantage of weeds destruction in cultural control? 3

1. **2-** The **destruction of vegetable trash and weeds**, frequently called **clean culture**, is also an important factor, especially in the control of hibernating insects
2. Weeds not only interfere with successful crop growth by **crowding**, but they may reduce the vigor of the crop by **competing for plant food** in the soil
3. In winter they provide **hiding places for hibernating pests**
4. A pest appearing in the spring before its crop food plant is available may find certain weeds palatable until the crop has appeared
5. Other pests move to late summer and fall weeds after their particular crops have been harvested

It largely involves manipulating farming practices to suppress **weed** growth and production, while promoting the development of the desired plant. The principles and techniques used to prevent **weed** spread are relevant to **cultural control**methods to limit the spread of **weeds** between different land areas.

13 Time of planting of sugarcane for cultural control? 5

IPM for Sugarcane borers(Cultural Control)

- **Time of planting**
- Early planting during December –January to escape the shoot borer incidence.
- Detrashing
- Internode borer and Stalk borer
- Manuring
- Avoid excess use nitrogenus fertilizers.
- Stalk borer and internode borer.

- Trash mulching and earthing up
- Removal and destruction of infested cane
- Early shoot borer
- Collection and destruction of eggs
- Internode borer and Top borer

14 Explain Biological control?5

1. **Parasitic and predaceous** insects as well as diseases of insects have already been discussed
2. As vital factors in the **natural control** of injurious insects
3. It is possible, however, for man to encourage such beneficial forms by introducing them into new areas
4. By growing them artificially for liberation whenever they become scarce in nature
5. Until recent years most of this work dealt with the introduction of parasites
6. From abroad **to combat those foreign insect pests** brought to this country accidentally but without their natural enemies
7. Certain **native parasites and predators** are now being reared in large numbers for liberation in areas badly infested with their host species
8. Probably the first introduction attempted from a foreign country was the now classic case of a **ladybeetle brought from Australia** to attack the **cottony cushion scale**
9. Which had already reached California and was endangering the existence of the citrus industry
10. Following this successful attempt many other parasites and predators have been imported
11. Which in many instances have proved valuable in reducing the numbers of foreign pests
12. Parasites should not be expected to kill off all their hosts
13. They will merely reduce the population to greater or less extent
14. Other control measures must also be used if the insect is to be exterminated
15. The discovery and introduction of parasites are expensive
16. However, since a successful parasite perpetuates itself, the initial high cost of establishment may be spread over a period of many years
17. Many dangers are involved in bringing parasites into a new country
18. These insects may attack not only the pest that is to be controlled
19. But also other parasites of the pest
20. In such cases the introduction may be of little value

21. There is always the chance also that their own secondary parasites may be imported along with the primary ones
22. Only thoroughly trained entomologists
23. Therefore, should be permitted to bring parasites and predators to this country if we are to avoid the introduction of undesirable insects along with the desirable ones
24. The liberation of insect diseases is not so generally successful as variations in natural conditions limit the growth of the disease
25. Many of the disease organisms are widely distributed and appear spontaneously
26. When suitable conditions prevail, even without any encouragement
27. The most systematic use of diseases in artificial insect control in the United States appears to be the annual distribution by the state of Florida of **fungi** that live on the **whiteflies that attack citrus**

15 Economic importance of insects? 5

A. Of the many thousands of insect species known to man some affect him and his property only slightly; some are beneficial; others are injurious in various degrees

B. Only rough estimates can be made of the monetary losses and gains by insect activities and even these are made with difficulty

C. Although it is the general practice to express losses in terms of dollar valuation

D. It probably would be safer to give them in bushels, tons, etc., because the relative values of different crops and products change from year to year

E. No crop of either vegetable or animal origin appears to be entirely free from insect injury

F. Field, truck and fruit of crops, both growing and in storage, household goods and food products, forests and the wood products derived from them, domestic animals and animal products, all are more or less liable to insect attack

G. Because there never has been a season free from the ravages of insects which might serve as a standard for comparison, much of this loss is not appreciated

H. If there could once be such a year entirely insect free, the difference would be apparent at once

16 How shot and top and internode borer can be biologically controlled? 10

Biological Control

- **For Early shoot borer :**
- *Trichogramma chilonis* ▶ **For Top borer :**
- *Trichogramma chilonis or T. japonicum, Isotima javensis* ▶ **For Internode borer :**

- *Trichogramma chilonis*
- Cultivation of resistance variety
- Irrigation at closer intervals for managing
- Early shoot borer
- Practice deep harvesting to destroy stubbles
- Removal of water shoots to destroy
- Top shoot borer

17 Sanitary control 3

What is Sanitary Control?

- **Methods to avoid introducing a pest into a field by**
- 1- Cleaning field equipment
- 2- Planting certified seed
- 3- Quarantines

18 Economically importance of insects 2

- The benefits to man from their activities are frequently overlooked in summing up the economic importance of insects
- Although in many cases it is impossible to say how much good is derived from insects

1-Attack on injurious species

- Reliable statistics are easily obtained relative to the production of useful insect products
- Those insects which attack injurious species probably form the most important group ▶

They are divided into two sections:

- 1-The parasites that live in or on the different stages of other insects, then called hosts
- 2-The predators that capture and devour other insects
- Parasites usually, though not always, live in or on a single host insect during their entire development
- Whereas a single predator may derive its nourishment from many victim insects.
- Parasites are of many species, frequently so specialized in their habits that one kind of parasite attacks only one or a few host species
- In some cases the parasites work only in insect eggs and thus prevent any feeding loss by the host insect
- Other parasites attack early stages of the pests, in some instances killing them soon

- In others not until the pests have completed their feeding in spite of being parasitized.
- In the latter the benefit is less, but at least there are fewer adults to produce the next generation
- All stages are known to be attacked in different species by one parasite or another

Pollination

- Some plants, as the Smyrna fig, clovers, melons, tomatoes, beans, peas and most of the common fruits, require the visits of insects before the seeds or fruits can form
- Another highly important group of insects includes those which pollinate flowers during their visits for nectar and pollen, thus enabling the plants to produce seeds
- The value of these crops which require the services of insects in the United States amounts to over two billion dollars

3-Commercial Products

- A third group of insects is responsible for a number of commercial products
- **1-Silk**
- **2-Honey**
- **3-Beeswax**
- **4-Shellac**

4-Diet of certain birds, mammals

- Insects provide the entire diet of certain birds, mammals, etc., and an important part of the diet for many others

5-Serve as Food

- Even man in some parts of the world consumes large quantities of grasshoppers, crickets, ants and other insects
- The great value of some insects and the great losses, sometimes enormous, caused by others make it obviously important to encourage the beneficial forms while striving to prevent or at least to reduce the losses caused by the injurious ones

19 Advantages of weeds destruction 3

6. **4- The destruction of weeds and brush** along fences is of advantage in the reduction of insects
7. But it results in the elimination of nesting and hiding places for insectivorous and game birds

20 Biological control 5

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21 Order of termites 2

Order: Blattodea

Isoptera

- **Includes:** Dry-wood termite, Subterranean termite
- **Mouthparts:** Chewing Mouthparts ▶ **Metamorphosis:** Incomplete Metamorphosis
- Live for 15 years.
- Lay 1 egg every 15 seconds
- Eat wood. Can destroy a house in 2-3 years
- Found in every US state except Alaska.
- Recycle wood in the soil

22 Pests of rice 10

PESTS OF RICE

Rice (*Oryza sativa* L.) is the most important staple food crop of the world

- More than 60 % of worlds population depend on it for food.

Major insect pests

- **1-Brown Plant Hopper**, *Nilaparvata lugens* (Homoptera: Delphacidae)
- **2-Yellow Rice Stem Borer**, *Scirpophaga incertulas* (Lepidoptera: Pyralidae)
- **3-Rice Ear-head Bug**, *Leptocorisa oratorius* (Hemiptera: Alydidae)
- **4-Rice Hispa,** *Dicladispa armigera (Coleoptera:* Chrysomelidae)
- **5-Rice Grasshopper**, *Hieroglyphus banian* (Orthoptera: Acrididae)

23 Family/ Order Symptoms of damage of Tobacco Cutworm 5 marks

Armyworm /Tobacco Cutworm: *Spodoptera litura* ▶

Family: Noctuidae ▶ **Order:** Lepidoptera.

- It is a major and important chewing insect pest of cotton crop.

Symptoms of damage

- Scrapping the epidermal layer, leaving the skeleton of veins of leaf
- During severe attack, only the stem and side shoots will be standing in the field without any leaf or bolls
- Larvae feed the leaves by making small holes.

24 Low and high temperature in artificial control

- Heat

2- Most insects cease reproduction and soon die at temperatures rom **100 to 1l0°F**

3- The temperature of their surroundings should be raised to **120° or higher** to kill them with heat artificially **within a few hours**

4- The length of period for effective treatment is more dependent upon the

   a. time to heat the grain

   b. clothing

   c. other material in which the insects are located

5- Than upon the heat resistance of the pests themselves

6- Because no insect can survive more than an hour if actually exposed to a temperature of

120°F 7- Bales and bags are difficult to heat through

cold

1. Although **low temperatures** are not so effective as high ones in killing insects
2. Many insect pests of stored products such as **furniture, rugs, clothing and seeds** become inactive at **40 to 45°F**.
3. Under these conditions infested materials suffer no further damage or increase in infestation
4. And materials free from insects when put into cold storage will remain so.
5. Those insects which hibernate outdoors withstand temperatures far below zero
6. Most insect pests of stored products, however, are unable to become dormant in the true sense and are killed in a relatively short time even at temperatures considerably above zero
7. The use of low temperatures in the artificial control of insects, there fore, is of two types
8. The more important is cold storage, as for furs, at temperatures just low enough to protect the stored materials from injury
9. The other use of cold actually to kill insects is feasible only on a large scale in zero weather in the north
10. Here as in the case of the heat-treatment much more time is required to chill grain, flour, clothing and other products than is usually realized

25 Fuzz purposes

Fuzz: It is used in production of mattresses, surgical cotton, photographic film and paper.

26 Scientific name of early shoot borer

- **For Early shoot borer :**
- *Trichogramma chilonis*

What is Biological Control?

- **Manipulation of biological organism to control pests by:**
- 1- Release of predators/parasites/disease of an insect or weed Can be time consuming, expensive and difficult **What is Host Plant Resistance?**
- Manipulating the crop to withstand or tolerate pests by
- 1- Natural breeding method
- 2- Genetically modified plants
- **Examples**: Glandular-haired Alfalfa, Bt Corn,

. where Rice Grass hopper live 2

- Various species of grasshopper are widely distributed in Nepal
- They are polyphagous and feed on leaves of rice, maize, millets, sugarcane, grasses etc

Which machinery is used to control pest 2

- Uses machinery and/or other tools to control pests
- 1- Tillage: Tillage is the agricultural preparation of soil by mechanical agitation of various types, such as digging, stirring, and overturning

Symptoms of damage of **Fruit borer** 3

- Bolls showing regular, circular bore holes
- Larvae seen feeding on the boll by thrusting their heads alone inside and leaving the rest of the body outside
- Presence of granular faecal pellets outside the bore hole.
- A single larva can damage 30-40 bolls.

Why Agronomic practices that are designed to: 3

- 1-Optimize growing conditions for the crop
- 2-Anything that increases a crop's competitive edge will result in increased tolerance to pests often resulting in reduced pesticide use

3- Create unfavorable conditions for the pest

1- Describe the places where heat treatment is used to kill the insects 3

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